Mr Dimelow

Current legislation:

Health and Safety

Topic	What students need to learn:
7 Safe working practices, potential hazards and risk assessment	7.1 - Adopting safe working practices, recognise and react to potential hazards: a) understanding safe working practices for yourself and others when designing and making, including when selecting and safely using machinery, equipment and tools in order to ensure safe working environments b) understanding the need for risk assessments – identification of potential hazards, identification of people at risk, evaluation of risks, implement control measures, recording and storing of risk assessment documentation

Topic	What students need to learn:
10 Current legislation	10.2 - The principles and applications of health and safety laws and regulations and their impact on the designing and making process, including the consequences of non-adherence: a) health and safety regulation – the Health and Safety Executive and an awareness of relevant regulations to manufacturing industries b) Health and Safety at Work etc Act (1974) – the procedures to safeguard the risk of injury to people: personal protective equipment (PPE), signage, warning symbols c) Control of Substances Hazardous to Health (COSHH) regulations – the storage and use of solvent-based substances containing volatile organic compounds (VOCs).

Starter - Video

- Write down all the health and Safety issues highlighted in the video
- Would you report Joe?
- How many unsafe Acts are there a year?
- How many near misses are there a year?
- What happened to Joe because of his actions?
- Why is Joe at fault and not his employer?

Starter - Video



Prevention

- Write down what could have Joe done differently? What precautions could Joe taken?

What is an ACCIDENT?

An unplanned event which may or may not result in an injury to a person or a loss to the company or a combination of both.

What is a HAZARD?

There are many definitions for hazard but the more common definition when talking about workplace health and safety is:

- A ***hazard*** is any source of potential damage, harm or adverse health effects on something or someone under certain conditions at work.
- Basically, a hazard can cause harm or adverse effects (to individuals as health effects or to organizations as property or equipment losses).

Ok so… what is a RISK?

- A ***risk*** is the likelihood that harm will occur, and a measure of the severity of the harm that the hazard poses.

What types of hazards are there?

- A common way to classify hazards is by category:
- **biological** - bacteria, viruses, insects, plants, birds, animals, and humans, etc.,
- **chemical** - depends on the physical, chemical and toxic properties of the chemical.
- **ergonomic** - repetitive movements, improper set up of workstation, etc.,
- **physical** - radiation, magnetic fields, pressure extremes (high pressure or vacuum), noise, etc,
- **psychological** - stress, violence, etc.,
- **safety** - slipping/tripping hazards, inappropriate machine guarding, equipment malfunctions or breakdowns

Why does ‘Health and Safety’ matter?

- In pairs write down as many possible hazards in the DT (engineering) workshop, list at least 10 hazard.

- Think about the possible effects on you from working in this environment.

Why does ‘Health and Safety’ matter?

- Possible hazards in the DT workshop, list at least 10 hazard.

Safe Working

There are ***three*** reasons why safe working is important in an engineering workplace:

1. **Cost** – *accidents can be expensive for employers*
2. **Morality** – *employers should look after their staff*
3. **The Law** – *there is a legal duty on everyone to work safely*

UK health and safety legislation

You have probably heard about the ***industrial revolution*** – when workers including children – worked long hours in cramped conditions for inadequate wages.

Whether we work in a factory, an office or at home – our working lives are governed by laws passed by parliament.








Essex Record Office
D/F 269/1/2683
2012 . Marconis Wireless Telegraph Works, Chelmsford. Machine Shop.

UK health and safety legislation

These are designed to protect the workforce from excessive demands by employers.

They also protect the employer against wrongful acts by their employees.

Employment legislation

In the UK – the principal rights and obligations imposed on employers and employees come from:

- Civil/Common law – covering contracts of employment between employer and employee
- UK Legislation
- European Legislation – which is then adopted by the UK Parliament

What is the difference between an Act and a Regulation??

Definition: An **Act** is a law that has been introduced in the Legislative Assembly as a Bill, has passed three readings and committee-study by the Legislative Assembly, and received Royal Assent. A **Regulation** is "subordinate legislation" authorised by an **Act** and made by Cabinet, a Minister, or a board or other body.

What is the difference between an Act and a Regulation??

- ***Acts*** – ***'statutory'*** – this means they have been passed by an Act of Parliament – it is a criminal offence to disobey them – they are the LAW *(Criminal Law)*
- ***Regulations*** – are ***'not statutory'*** – but they amplify the law

The main legal instrument for ensuring safe workplaces today is the

'Health and Safety at Work Act 1974'

Health and Safety at Work Act 1974

- The **HASAWA** is a UK Act of Parliament – anyone who does not comply with this Act is breaking the law
- The Act is enforced by the **Health and Safety Executive (HSE)**

Health and Safety Executive (HSE)

Health and Safety Executive (HSE)

- The HSE is the enforcing authority for workplace safety under powers set out in the HASAWA.
- The HSE appoints inspectors to monitor H&S in workplaces.
- These inspectors have wide-ranging powers to ensure that engineering companies comply with legislation.
- Their role is to ensure that employees and others are protected, to enforce the law and to advise on matters related to H&S.

Section 2 of HASAWA 1974

- This formalises the legal responsibilities of employers in respect to their employees

Video HASAWA 1974



Main elements of the HASAWA 1974 (See Slides 25 – 28)

EMPLOYERS DUTIES

2(1)

To ensure as far as is reasonably practicable the health, safety and welfare of all his employees

2(2)

a) Provide and maintain safe plant and equipment – safe systems of work

b) Safe transport and storage of materials and substances including handling

c) Provide information, instruction, training and supervision

d) Provide and maintain safe place of work without risks to health, including access and egress

e) Provide and maintain a safe working environment – without risk to health

Prepare a written health & safety policy and bring it to the notice of all employees

To Others

3

1) Duty of care for own safety and safety of others (visitors, trespassers)

2) Duty of self employed persons to have a duty of care to others

3) Each shall provide information of hazards etc. which may affect others

4

1) Duty of care to others using the same premises

2) Ensure plant, equipment, substances etc. are safe without risk to health, including access and egress

5

Concerns the control of pollution by best practicable means

EMPLOYEES DUTIES

7

a) Care for own safety and the safety of others who may be affected by actions or omissions

b) Co-operate with employer in discharging statutory obligations

8

Not to interfere with or misuse equipment provided for safety

9

Must not charge employees for safety clothing and equipment

6

To promote intrinsic safety in articles and substances used at work, so by requiring suppliers, manufacturers and designers (importers and hire firms) to provide articles and substances which do not give rise to risks to health and safety

Employers Duties

Section 2 of the HASAWA formalises the main responsibilities of employers:

- To ensure as far as is reasonably practicable the health, safety and welfare of all his employees – they have a ***duty of care*** to their employees
- Provide and maintain a safe place of work without risks to health
- Provide training and supervision
- Safe transport and storage of materials and substances

Employees Duties

Section 7 of the HASAWA formalises the two main responsibilities of employees:

1. To take reasonable care of their own H&S and others who may be affected by their acts
2. To co-operate with their employer

Employees – General Rules

1. Be aware of your responsibility under section 7 of the HASAWA
2. Be aware of the hazards and the risks from any engineering activity in which you will be engaged.
3. Be familiar with the risk assessments that have been made.
4. Make sure you know about any relevant regulations.

Health and Safety Regulations

The HASAWA 1974 is ***'amplified'*** by health and safety ***'regulations'*** – these are ***NOT*** **STATUTORY** but work with the Act (law)

There are lots of health and safety 'regulations'. These can be selected in accordance with business organisation and/or the type of processes being carried out.

Health and Safety Regulations

For the A level PD specification you need to be familiar with the following health and safety 'regulations':

- *Management of Health and Safety at Work Regulations – **'Risk Assessments'***
- *Control of Substances Hazardous to Health (COSHH) regulations – including warning symbols*
- *Personal protective equipment (PPE) Regulations*
- *Safety Signs and Signals Regulations*